Class 12 Mathematics

Chapter 2: Inverse Trigonometric Functions

Practice Set B: step-wise solutions

20 questions **43** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** NCERT Exemplar — 1 mark

The value of $\sin^{-1}\left(\cos\frac{33\pi}{5}\right)$ is:

(A) $\frac{3\pi}{5}$ **(B)** $-\frac{7\pi}{5}$ **(C)** $\frac{\pi}{10}$ **(D)** $-\frac{\pi}{10}$

Formula / idea: Turn the cosine into the sine of an angle in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, the range of $\sin^{-1}x$.

1 Remove whole turns of 2π.
$$\cos\frac{33\pi}{5}=\cos\left(6\pi+\frac{3\pi}{5}\right)=\cos\frac{3\pi}{5}$$

2 Write the cosine as a sine, using $\cos\theta=\sin\left(\frac{\pi}{2}-\theta\right)$.
$$\cos\frac{3\pi}{5}=\sin\left(\frac{\pi}{2}-\frac{3\pi}{5}\right)=\sin\left(-\frac{\pi}{10}\right)$$

3 $-\frac{\pi}{10}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, so $\sin^{-1}$ and $\sin$ cancel. ✓ 1
$$\sin^{-1}\left(\sin\left(-\frac{\pi}{10}\right)\right)=-\frac{\pi}{10}$$

4 Hence the value is $-\frac{\pi}{10}$. Option (D).

Answer **(D)** $-\frac{\pi}{10}$

Watch out: Writing $\frac{3\pi}{5}$: $\sin^{-1}$ does not undo $\cos$, and $\frac{3\pi}{5}$ is outside $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

2 MCQ Board-pattern 1 mark

The value of $\tan^{-1}\left(\cot\frac{7\pi}{5}\right)$ is:

(A) $\frac{7\pi}{5}$ **(B)** $\frac{2\pi}{5}$ **(C)** $-\frac{9\pi}{10}$ **(D)** $\frac{\pi}{10}$

Formula / idea: $\tan^{-1}(\tan\theta)=\theta$ only when $\theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

1 $\cot$ has period π.
$$\cot\frac{7\pi}{5}=\cot\left(\pi+\frac{2\pi}{5}\right)=\cot\frac{2\pi}{5}$$

2 Write the cot as a tan, using $\cot\theta=\tan\left(\frac{\pi}{2}-\theta\right)$.
$$\cot\frac{2\pi}{5}=\tan\left(\frac{\pi}{2}-\frac{2\pi}{5}\right)=\tan\frac{\pi}{10}$$

3 $\frac{\pi}{10}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, so $\tan^{-1}$ and $\tan$ cancel. ✓ 1
$$\tan^{-1}\left(\tan\frac{\pi}{10}\right)=\frac{\pi}{10}$$

4 Hence the value is $\frac{\pi}{10}$. Option (D).

Answer **(D)** $\frac{\pi}{10}$

Watch out: Choosing $\frac{2\pi}{5}$: $\tan^{-1}$ does not undo $\cot$; change $\cot$ to $\tan$ first.

3 MCQ NCERT Exemplar 1 mark

The domain of the function defined by $f(x)=\sin^{-1}\sqrt{x-1}$ is:

(A) $[1,2]$ **(B)** $[-1,1]$ **(C)** $[0,1]$ **(D)** none of these

Formula / idea: Both conditions must hold: the root needs $x-1\geq 0$, and $\sin^{-1}t$ needs $-1\leq t\leq 1$.

1 The square root is defined only when $x-1\geq 0$, and then it is never negative.
$$\sqrt{x-1}\geq 0$$

2 $\sin^{-1}$ needs its input in $[-1,1]$; with the root ≥ 0:
$$0\leq\sqrt{x-1}\leq 1$$

3 Square each part. ✓ 1
$$0\leq x-1\leq 1\ \Rightarrow\ 1\leq x\leq 2$$

4 Hence the domain is $[1, 2]$. Option (A).

Answer **(A)** $[1, 2]$

Watch out: Choosing $[-1, 1]$: that is the domain of $\sin^{-1} t$, not of $f(x)$.

4 MCQ Board-pattern 1 mark

If $x \leq -1$, then the principal value $\sec^{-1} x$ lies in the interval:

(A) $\left[-\frac{\pi}{2}, 0\right)$ **(B)** $\left(\frac{\pi}{2}, \pi\right]$ **(C)** $\left[0, \frac{\pi}{2}\right)$ **(D)** $\left[\pi, \frac{3\pi}{2}\right)$

Formula / idea: Principal value branch of $\sec^{-1} x$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$.

1 Let $\sec^{-1} x = \theta$.
$\sec\theta = x, \qquad \theta \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$

2 Since $x \leq -1$, the cosine is negative.
$\cos\theta = \frac{1}{x} \in [-1, 0)$

3 In $[0, \pi]$, $\cos\theta < 0$ only for $\theta > \frac{\pi}{2}$; $\cos\theta = -1$ at $\theta = \pi$. ✓ 1
$\theta \in \left(\frac{\pi}{2}, \pi\right]$

4 Hence $\sec^{-1} x$ lies in $\left(\frac{\pi}{2}, \pi\right]$. Option (B).

Answer **(B)** $\left(\frac{\pi}{2}, \pi\right]$

Watch out: Choosing $\left[-\frac{\pi}{2}, 0\right)$: a negative input does not give a negative angle; $\sec^{-1} x$ is never negative.

5 MCQ Board-pattern 1 mark

The value of $\sec^{-1}\left(-\sqrt{2}\right) - \operatorname{cosec}^{-1}\left(-\frac{2}{\sqrt{3}}\right)$ is:

(A) $\frac{5\pi}{12}$ **(B)** $\frac{13\pi}{12}$ **(C)** $\frac{\pi}{12}$ **(D)** $-\frac{7\pi}{12}$

Formula / idea: Ranges: $\sec^{-1} x \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$, $\operatorname{cosec}^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$.

1 $\sec\frac{3\pi}{4} = -\sqrt{2}$ and $\frac{3\pi}{4}$ is in the range of $\sec^{-1}$.
$\sec^{-1}\left(-\sqrt{2}\right) = \frac{3\pi}{4}$

2 $\operatorname{cosec}\left(-\frac{\pi}{3}\right) = -\frac{2}{\sqrt{3}}$ and $-\frac{\pi}{3}$ is in the range of $\operatorname{cosec}^{-1}$.

$\operatorname{cosec}^{-1}\left(-\frac{2}{\sqrt{3}}\right) = -\frac{\pi}{3}$

3 Subtract. ✓ 1

$\frac{3\pi}{4} - \left(-\frac{\pi}{3}\right) = \frac{9\pi + 4\pi}{12} = \frac{13\pi}{12}$

4 Hence the value is $\frac{13\pi}{12}$. Option (B).

Answer **(B)** $\frac{13\pi}{12}$

Watch out: Taking $\sec^{-1}\left(-\sqrt{2}\right) = -\frac{\pi}{4}$, which gives $\frac{\pi}{12}$; $-\frac{\pi}{4}$ is outside the range of $\sec^{-1}$.

6 **Assertion-Reason** CBSE SQP 2022-23 1 mark

Assertion (A): The domain of the function $\sec^{-1} 2x$ is $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$. **Reason (R):** $\sec^{-1}(-2) = -\frac{\pi}{4}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $\sec^{-1} t$ is defined for $t \le -1$ or $t \ge 1$; its values lie in $[0, \pi] - \left\{\frac{\pi}{2}\right\}$.

1 Check A: $2x$ must lie outside $(-1, 1)$, so A is true.

$2x \le -1$ or $2x \ge 1 \Rightarrow x \le -\frac{1}{2}$ or $x \ge \frac{1}{2}$

2 Check R: $\sec\frac{2\pi}{3} = -2$ and $\frac{2\pi}{3} \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$, so R is false.

$\sec^{-1}(-2) = \frac{2\pi}{3} \ne -\frac{\pi}{4}$

3 R is false, so it cannot explain A.

4 A is true and R is false. Option (C). ✓ 1

Answer **(C) Assertion (A) is true, but Reason (R) is false.**

Watch out: Accepting $-\frac{\pi}{4}$: it is not in the range of $\sec^{-1}$, and $\sec\left(-\frac{\pi}{4}\right) = \sqrt{2}$, not -2.

7 **Short answer** PYQ-style 2 marks

Find the value of $\sin^{-1}\left(\sin\frac{13\pi}{7}\right)+\cos^{-1}\left(\cos\frac{13\pi}{7}\right)$.

Formula / idea: Bring each angle into its own principal range: $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ for $\sin^{-1}$, $[0,\pi]$ for $\cos^{-1}$.

1 Write the angle as 2π minus a small angle.
$\frac{13\pi}{7}=2\pi-\frac{\pi}{7}$

2 Use $\sin(2\pi-\theta)=-\sin\theta=\sin(-\theta)$.
$\sin\frac{13\pi}{7}=\sin\left(-\frac{\pi}{7}\right)$

3 $-\frac{\pi}{7}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, so:
$\sin^{-1}\left(\sin\frac{13\pi}{7}\right)=-\frac{\pi}{7}$ ✓ 1

4 Use $\cos(2\pi-\theta)=\cos\theta$; $\frac{\pi}{7}\in[0,\pi]$.
$\cos^{-1}\left(\cos\frac{13\pi}{7}\right)=\cos^{-1}\left(\cos\frac{\pi}{7}\right)=\frac{\pi}{7}$ ✓ ½

5 Add the two values.
$-\frac{\pi}{7}+\frac{\pi}{7}=0$ ✓ ½

Answer 0

Watch out: Using the same answer for both terms: $\sin^{-1}$ and $\cos^{-1}$ have different principal ranges.

Level 2

16 marks

8 **Short answer** Board-pattern 2 marks

Evaluate: $\tan^{-1}\left(2\cos\frac{2\pi}{3}\right)+\sec^{-1}\left(\operatorname{cosec}\left(-\frac{\pi}{6}\right)\right)+\cot^{-1}\left(\tan\left(-\frac{\pi}{3}\right)\right)$.

Formula / idea: Work out each inner value, then take the principal value in the right range.

1 $2\cos\frac{2\pi}{3}=-1$; $-\frac{\pi}{4}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.
$\tan^{-1}(-1)=-\frac{\pi}{4}$ ✓ ½

2 $\operatorname{cosec}\left(-\frac{\pi}{6}\right)=-2$; $\sec\frac{2\pi}{3}=-2$ with $\frac{2\pi}{3}\in[0,\pi]-\left\{\frac{\pi}{2}\right\}$.
$\sec^{-1}(-2)=\frac{2\pi}{3}$ ✓ ½

3 $\tan\left(-\frac{\pi}{3}\right) = -\sqrt{3}$; $\cot\frac{5\pi}{6} = -\sqrt{3}$ with $\frac{5\pi}{6} \in (0, \pi)$.
$\cot^{-1}\left(-\sqrt{3}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ ✓ ½

4 Add, using the LCM 12. ✓ ½
$-\frac{\pi}{4} + \frac{2\pi}{3} + \frac{5\pi}{6} = \frac{-3\pi + 8\pi + 10\pi}{12} = \frac{15\pi}{12} = \frac{5\pi}{4}$

Answer $\frac{5\pi}{4}$

Watch out: Taking $\cot^{-1}\left(-\sqrt{3}\right) = -\frac{\pi}{6}$; the range of $\cot^{-1}$ is $(0, \pi)$.

9 **Short answer** Board-pattern 2 marks

Find the domain of $f(x) = \sin^{-1}\left(x^2 - 2x\right)$.

Formula / idea: $\sin^{-1} t$ is defined only for $-1 \le t \le 1$; solve both inequalities.

1 The argument must lie in $[-1, 1]$. ✓ ½
$-1 \le x^2 - 2x \le 1$

2 Left part: it is a perfect square, true for every real x. ✓ ½
$x^2 - 2x + 1 \ge 0 \;\Leftrightarrow\; (x-1)^2 \ge 0$

3 Right part: find the roots of $x^2 - 2x - 1 = 0$.
$x^2 - 2x - 1 \le 0, \qquad x = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$

4 The quadratic is ≤ 0 between its roots. ✓ ½
$1 - \sqrt{2} \le x \le 1 + \sqrt{2}$

5 Both parts hold together, so: ✓ ½
$\text{Domain} = \left[1 - \sqrt{2},\ 1 + \sqrt{2}\right]$

Answer **Domain** $= \left[1 - \sqrt{2},\ 1 + \sqrt{2}\right]$

Watch out: Writing $x \le 1 - \sqrt{2}$ or $x \ge 1 + \sqrt{2}$; the quadratic is negative between its roots, not outside.

10 **Short answer** Board-pattern 2 marks

Find the domain of $f(x) = \sin^{-1}(2x - 1) + \text{cosec}^{-1}(3x)$.

Formula / idea: A sum is defined where both terms are defined: intersect the two domains.

1 For $\sin^{-1}(2x - 1)$, the argument lies in $[-1, 1]$. ✓ ½
$-1 \le 2x - 1 \le 1 \;\Rightarrow\; 0 \le x \le 1$

2 For $\operatorname{cosec}^{-1}(3x)$, the argument lies outside $(-1, 1)$. ✓ ½

$3x \le -1$ or $3x \ge 1 \Rightarrow x \le -\frac{1}{3}$ or $x \ge \frac{1}{3}$

3 Take the common part. ✓ ½

$[0,1] \cap \left(\left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{1}{3}, \infty\right)\right) = \left[\frac{1}{3}, 1\right]$

4 Hence the domain of f: ✓ ½

$\text{Domain} = \left[\frac{1}{3}, 1\right]$

Answer **Domain** $= \left[\frac{1}{3}, 1\right]$

Watch out: Using $-1 \le 3x \le 1$ for $\operatorname{cosec}^{-1}$; that is exactly where it is not defined.

11 **Short answer** Board-pattern 2 marks

Find the value of $\cos^{-1}\left(\cos\frac{5\pi}{4}\right) + \cot^{-1}\left(\cot\left(-\frac{\pi}{4}\right)\right)$.

Formula / idea: $\cos^{-1}(\cos\theta) = \theta$ for $\theta \in [0, \pi]$; $\cot^{-1}(\cot\theta) = \theta$ for $\theta \in (0, \pi)$.

1 Use $\cos(2\pi - \theta) = \cos\theta$. ✓ ½

$\cos\frac{5\pi}{4} = \cos\left(2\pi - \frac{5\pi}{4}\right) = \cos\frac{3\pi}{4}$

2 $\frac{3\pi}{4} \in [0, \pi]$, so: ✓ ½

$\cos^{-1}\left(\cos\frac{5\pi}{4}\right) = \frac{3\pi}{4}$

3 $\cot$ has period π, and $\frac{3\pi}{4} \in (0, \pi)$. ✓ ½

$\cot^{-1}\left(\cot\left(-\frac{\pi}{4} + \pi\right)\right) = \cot^{-1}\left(\cot\frac{3\pi}{4}\right) = \frac{3\pi}{4}$

4 Add the two values. ✓ ½

$\frac{3\pi}{4} + \frac{3\pi}{4} = \frac{3\pi}{2}$

Answer $\frac{3\pi}{2}$

Watch out: Writing $\cot^{-1}\left(\cot\left(-\frac{\pi}{4}\right)\right) = -\frac{\pi}{4}$; that angle is outside $(0, \pi)$.

12 **Short answer** PYQ-style 2 marks

Evaluate: $\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right)$.

Formula / idea: $\tan^2\frac{\theta}{2} = \frac{1-\cos\theta}{1+\cos\theta}$

1 Let $\cos^{-1}\frac{\sqrt{5}}{3} = \theta$. The input is positive, so θ is acute. ✓ ½

$\cos\theta = \frac{\sqrt{5}}{3}, \qquad 0 < \theta < \frac{\pi}{2} \Rightarrow 0 < \frac{\theta}{2} < \frac{\pi}{4}$

2 Use the half-angle formula. ✓ ½

$\tan^2\frac{\theta}{2} = \frac{1-\frac{\sqrt{5}}{3}}{1+\frac{\sqrt{5}}{3}} = \frac{3-\sqrt{5}}{3+\sqrt{5}}$

3 Multiply top and bottom by $3-\sqrt{5}$. ✓ ½

$\tan^2\frac{\theta}{2} = \frac{(3-\sqrt{5})^2}{9-5} = \left(\frac{3-\sqrt{5}}{2}\right)^2$

4 $\frac{\theta}{2}$ is acute, so take the positive root. ✓ ½

$\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right) = \frac{3-\sqrt{5}}{2}$

Answer $\frac{3-\sqrt{5}}{2}$

Watch out: Halving the number instead of the angle, i.e. working with $\cos^{-1}\frac{\sqrt{5}}{6}$.

13 **Short answer** PYQ-style 3 marks

If $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$, find the value of x.

Formula / idea: Find the interval of the angle; $\sin\theta = 1$ there only at $\theta = \frac{\pi}{2}$.

1 Let $\sin^{-1}\frac{1}{5} = \alpha$. The input is positive, so α is acute. ✓ ½

$\sin\alpha = \frac{1}{5}, \qquad 0 < \alpha < \frac{\pi}{2}$

2 $\cos^{-1}x \in [0, \pi]$, so the whole angle lies in: ✓ ½

$0 < \alpha + \cos^{-1}x < \frac{3\pi}{2}$

3 In $\left(0, \frac{3\pi}{2}\right)$, sine equals 1 only at $\frac{\pi}{2}$. ✓ 1

$\alpha + \cos^{-1} x = \frac{\pi}{2}$

4 Make $\cos^{-1} x$ the subject. ✓ ½

$\cos^{-1} x = \frac{\pi}{2} - \alpha$

5 Take cosine of both sides. ✓ ½

$x = \cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha = \frac{1}{5}$

Answer $x = \frac{1}{5}$

Watch out: Writing $x = \cos\frac{1}{5}$: the cosine is of the angle $\frac{\pi}{2} - \alpha$, not of the number $\frac{1}{5}$.

14 **Short answer** PYQ-style 3 marks

Evaluate: $\cos\left(\sin^{-1}\frac{1}{4} + \sec^{-1}\frac{4}{3}\right)$.

Formula / idea: $\cos(A+B) = \cos A\cos B - \sin A\sin B$

1 Let $\sin^{-1}\frac{1}{4} = A$, an acute angle. ✓ ½

$\sin A = \frac{1}{4}, \qquad 0 < A < \frac{\pi}{2}$

2 Find $\cos A$; it is positive as A is acute. ✓ ½

$\cos A = \sqrt{1 - \frac{1}{16}} = \frac{\sqrt{15}}{4}$

3 Let $\sec^{-1}\frac{4}{3} = B$, also acute. ✓ ½

$\sec B = \frac{4}{3} \Rightarrow \cos B = \frac{3}{4}$

4 Find $\sin B$; it is positive as B is acute. ✓ ½

$\sin B = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$

5 Use the formula for $\cos(A+B)$. ✓ ½

$\cos(A+B) = \frac{\sqrt{15}}{4}\cdot\frac{3}{4} - \frac{1}{4}\cdot\frac{\sqrt{7}}{4}$

6 Simplify.

$$\cos\left(\sin^{-1}\frac{1}{4}+\sec^{-1}\frac{4}{3}\right)=\frac{3\sqrt{15}-\sqrt{7}}{16}$$

✓ ½

Answer $\dfrac{3\sqrt{15}-\sqrt{7}}{16}$

Watch out: Taking $\cos B=\frac{4}{3}$: secant is the reciprocal, so $\cos B=\frac{3}{4}$.

Level 3

19 marks

15 **Long answer** Board-pattern 3 marks

On the same axes, draw the graphs of $y=\sin^{-1}x$ and $y=\cos^{-1}x$ for their principal value branches. Using the graphs, find the point where they meet, and write the values of x for which $\sin^{-1}x>\cos^{-1}x$.

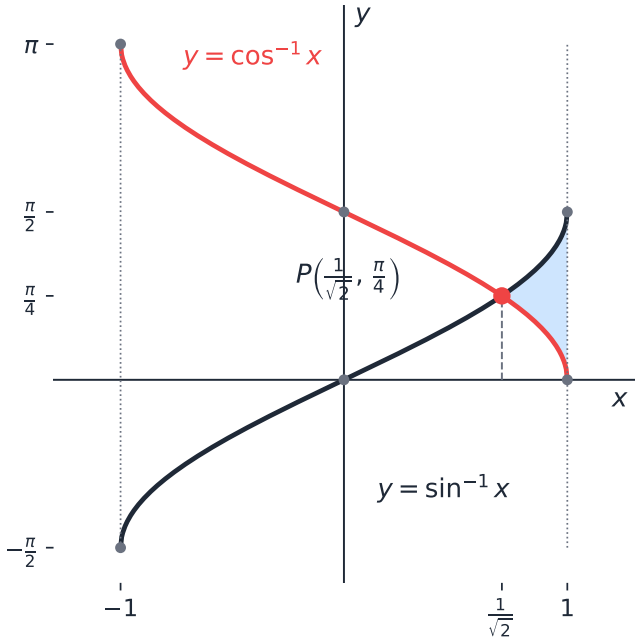
Formula / idea: On $[-1,1]$, $\sin^{-1}x$ increases from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$; $\cos^{-1}x$ decreases from π to 0.

1 Make a table of values for both functions.

x	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
$\sin^{-1}x$	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$
$\cos^{-1}x$	π	$\frac{2\pi}{3}$	$\frac{\pi}{2}$	$\frac{\pi}{3}$	0

✓ 1

2 Plot both sets of points on one axes and join each by a smooth curve.



✓ 1

3 At the meeting point let $\sin^{-1}x=\cos^{-1}x=\theta$; then $\sin\theta=\cos\theta$, $\theta\in\left[0,\frac{\pi}{2}\right]$.

$$\theta=\frac{\pi}{4},\quad x=\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}\ \Rightarrow\ P\left(\frac{1}{\sqrt{2}},\frac{\pi}{4}\right)$$

✓ ½

4 To the right of P the $\sin^{-1} x$ curve is above the $\cos^{-1} x$ curve (shaded).
$\sin^{-1} x > \cos^{-1} x \quad \text{for} \quad \frac{1}{\sqrt{2}} < x \leq 1$ ✓ ½

Answer **The graphs meet at** $\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right)$**;** $\sin^{-1} x > \cos^{-1} x$ **for** $\frac{1}{\sqrt{2}} < x \leq 1$**.**

Watch out: Including $x = \frac{1}{\sqrt{2}}$: there the two values are equal, not greater.

16 **Long answer** Board-pattern 3 marks

Find the value of $\tan^{-1}\left(\tan\frac{9\pi}{8}\right) + \sec^{-1}\left(\sec\frac{7\pi}{4}\right) + \operatorname{cosec}^{-1}\left(\operatorname{cosec}\frac{4\pi}{3}\right)$.

Formula / idea: Ranges: $\tan^{-1} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\sec^{-1} \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$, $\operatorname{cosec}^{-1} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$.

1 $\tan$ has period π.
$\tan\frac{9\pi}{8} = \tan\left(\pi + \frac{\pi}{8}\right) = \tan\frac{\pi}{8}$ ✓ ½

2 $\frac{\pi}{8} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so:
$\tan^{-1}\left(\tan\frac{9\pi}{8}\right) = \frac{\pi}{8}$ ✓ ½

3 Use $\sec(2\pi - \theta) = \sec\theta$.
$\sec\frac{7\pi}{4} = \sec\left(2\pi - \frac{\pi}{4}\right) = \sec\frac{\pi}{4}$ ✓ ½

4 $\frac{\pi}{4} \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$, so:
$\sec^{-1}\left(\sec\frac{7\pi}{4}\right) = \frac{\pi}{4}$ ✓ ½

5 $\operatorname{cosec}(\pi + \theta) = -\operatorname{cosec}\theta = \operatorname{cosec}(-\theta)$, and $-\frac{\pi}{3}$ is in the range.
$\operatorname{cosec}^{-1}\left(\operatorname{cosec}\frac{4\pi}{3}\right) = \operatorname{cosec}^{-1}\left(\operatorname{cosec}\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3}$ ✓ ½

6 Add, using the LCM 24.
$\frac{\pi}{8} + \frac{\pi}{4} - \frac{\pi}{3} = \frac{3\pi + 6\pi - 8\pi}{24} = \frac{\pi}{24}$ ✓ ½

Answer $\frac{\pi}{24}$

Watch out: Writing $\operatorname{cosec}^{-1}\left(\operatorname{cosec}\frac{4\pi}{3}\right) = \frac{\pi}{3}$ and losing the minus sign.

17 **Long answer** Board-pattern 3 marks

Find the domain and range of $f(x) = \cos^{-1}\left(\dfrac{x^2-1}{2}\right)$.

Formula / idea: $\cos^{-1} t$ needs $-1 \le t \le 1$ and is a decreasing function of t.

1 The argument must lie in $[-1, 1]$.
$$-1 \le \frac{x^2-1}{2} \le 1$$ ✓ ½

2 Multiply by 2, then add 1.
$$-2 \le x^2 - 1 \le 2 \;\Rightarrow\; -1 \le x^2 \le 3$$ ✓ ½

3 $x^2 \ge -1$ always holds; $x^2 \le 3$ gives:
$$-\sqrt{3} \le x \le \sqrt{3} \;\Rightarrow\; \text{Domain} = \left[-\sqrt{3}, \sqrt{3}\right]$$ ✓ ½

4 On this domain x^2 takes every value in $[0, 3]$, so the argument t takes:
$$t = \frac{x^2-1}{2} \in \left[-\frac{1}{2}, 1\right]$$ ✓ ½

5 $\cos^{-1}$ is decreasing, so find its values at the two ends.
$$\cos^{-1}(1) = 0, \qquad \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$ ✓ ½

6 Hence the range of f:
$$\text{Range} = \left[0, \frac{2\pi}{3}\right]$$ ✓ ½

Answer **Domain** $= \left[-\sqrt{3}, \sqrt{3}\right]$**; range** $= \left[0, \dfrac{2\pi}{3}\right]$

Watch out: Writing the range as $[0, \pi]$ by habit; here the argument never goes below $-\frac{1}{2}$.

18 **Long answer** NCERT Ex 2.2 Q7 3 marks

Write $\tan^{-1}\left(\dfrac{3a^2x - x^3}{a^3 - 3ax^2}\right)$, $a > 0$, $-\dfrac{a}{\sqrt{3}} < x < \dfrac{a}{\sqrt{3}}$, in the simplest form.

Formula / idea: Put $x = a\tan\theta$ and use $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$.

1 Put $x = a\tan\theta$.
$$\theta = \tan^{-1}\frac{x}{a}$$ ✓ ½

2 Find the interval of θ from the given interval of x.
$$-\frac{1}{\sqrt{3}} < \tan\theta < \frac{1}{\sqrt{3}} \;\Rightarrow\; -\frac{\pi}{6} < \theta < \frac{\pi}{6}$$ ✓ ½

3 Substitute and take a^3 common from top and bottom. ✓ ½
$$\frac{3a^3\tan\theta - a^3\tan^3\theta}{a^3 - 3a^3\tan^2\theta} = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

4 Recognise the triple-angle formula. ✓ ½
$$\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = \tan 3\theta$$

5 $-\frac{\pi}{2} < 3\theta < \frac{\pi}{2}$, so $\tan^{-1}$ and $\tan$ cancel. ✓ ½
$\tan^{-1}(\tan 3\theta) = 3\theta$

6 Put back $\theta = \tan^{-1}\frac{x}{a}$. ✓ ½
$$\tan^{-1}\left(\frac{3a^2x - x^3}{a^3 - 3ax^2}\right) = 3\tan^{-1}\frac{x}{a}$$

Answer $3\tan^{-1}\frac{x}{a}$

Watch out: Skipping the check $3\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; without it $\tan^{-1}(\tan 3\theta) = 3\theta$ is not justified.

19 **Long answer** NCERT Misc. Ex. Q9 3 marks

Prove that $\cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4}\right)$.

Formula / idea: $1 \pm \sin x = \left(\cos\frac{x}{2} \pm \sin\frac{x}{2}\right)^2$

1 Write $1 + \sin x$ as a perfect square. ✓ ½
$1 + \sin x = \cos^2\frac{x}{2} + \sin^2\frac{x}{2} + 2\sin\frac{x}{2}\cos\frac{x}{2} = \left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)^2$

2 Similarly for $1 - \sin x$. ✓ ½
$1 - \sin x = \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2$

3 $0 < \frac{x}{2} < \frac{\pi}{8}$, so $\cos\frac{x}{2} > \sin\frac{x}{2} > 0$. Take positive roots. ✓ ½
$\sqrt{1+\sin x} = \cos\frac{x}{2} + \sin\frac{x}{2}$, $\sqrt{1-\sin x} = \cos\frac{x}{2} - \sin\frac{x}{2}$

4 Add for the top, subtract for the bottom. ✓ ½
$$\frac{2\cos\frac{x}{2}}{2\sin\frac{x}{2}} = \cot\frac{x}{2}$$

5 $\frac{x}{2} \in \left(0, \frac{\pi}{8}\right) \subset (0, \pi)$, so $\cot^{-1}$ and $\cot$ cancel. ✓ ½
$\text{LHS} = \cot^{-1}\left(\cot\frac{x}{2}\right) = \frac{x}{2}$

6 Hence LHS = RHS.

$$\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right)=\frac{x}{2}$$

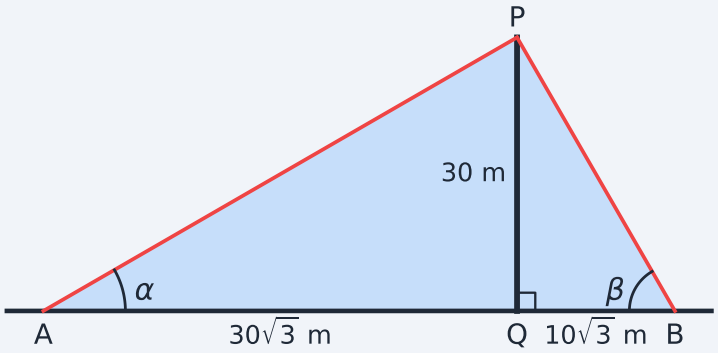
✓ ½

Answer **LHS** $=\cot^{-1}\left(\cot\frac{x}{2}\right)=\frac{x}{2}=$ **RHS. Hence proved.**

Watch out: Writing $\sqrt{1-\sin x}=\sin\frac{x}{2}-\cos\frac{x}{2}$: that is negative here, and a square root is never negative.

20 Case study Board-pattern — 4 marks

Case Study: A vertical tower PQ, 30 m tall, stands on level ground with top P and foot Q. Two observers A and B stand on opposite sides of the tower, in line with Q, with QA $=30\sqrt{3}$ m and QB $=10\sqrt{3}$ m. The angles of elevation of P from A and from B are $\alpha=\tan^{-1}\left(\frac{\text{PQ}}{\text{QA}}\right)$ and $\beta=\tan^{-1}\left(\frac{\text{PQ}}{\text{QB}}\right)$.



(i) Find α. — 1 mark

(ii) Find β, and hence the angle APB at the top of the tower. — 1 mark

(iii) Find the values of $\sin^{-1}[\sin(\alpha+2\beta)]$ and $\cos^{-1}[\cos(\alpha-2\beta)]$. — 2 marks

Formula / idea: In right triangle PQA, $\tan\alpha=\frac{\text{PQ}}{\text{QA}}$; the angles of triangle APB add up to π.

(i) — 1 mark

1 Put PQ $=30$, QA $=30\sqrt{3}$.

$$\alpha=\tan^{-1}\left(\frac{30}{30\sqrt{3}}\right)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

✓ ½

2 $\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}$ and $\frac{\pi}{6}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

$$\alpha=\frac{\pi}{6}$$

✓ ½

Answer (i) $\alpha=\frac{\pi}{6}$

(ii) 1 mark

1 Put PQ $= 30$, QB $= 10\sqrt{3}$; $\tan\dfrac{\pi}{3} = \sqrt{3}$. ✓ ½

$$\beta = \tan^{-1}\left(\frac{30}{10\sqrt{3}}\right) = \tan^{-1}\sqrt{3} = \frac{\pi}{3}$$

2 Angle sum of triangle APB. ✓ ½

$$\angle\text{APB} = \pi - (\alpha + \beta) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

Answer (ii) $\beta = \dfrac{\pi}{3}$; $\angle\text{APB} = \dfrac{\pi}{2}$

(iii) 2 marks

1 $\alpha + 2\beta = \dfrac{5\pi}{6}$ is outside $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$; use $\sin(\pi - \theta) = \sin\theta$. ✓ ½

$$\sin\frac{5\pi}{6} = \sin\left(\pi - \frac{5\pi}{6}\right) = \sin\frac{\pi}{6}$$

2 $\dfrac{\pi}{6} \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$, so: ✓ ½

$$\sin^{-1}\left[\sin(\alpha + 2\beta)\right] = \frac{\pi}{6}$$

3 $\alpha - 2\beta = -\dfrac{\pi}{2}$ is outside $[0, \pi]$; cosine is even. ✓ ½

$$\cos\left(-\frac{\pi}{2}\right) = \cos\frac{\pi}{2}$$

4 $\dfrac{\pi}{2} \in [0, \pi]$, so: ✓ ½

$$\cos^{-1}\left[\cos(\alpha - 2\beta)\right] = \frac{\pi}{2}$$

Answer (iii) $\sin^{-1}\left[\sin(\alpha + 2\beta)\right] = \dfrac{\pi}{6}$; $\cos^{-1}\left[\cos(\alpha - 2\beta)\right] = \dfrac{\pi}{2}$

Watch out: In (iii) writing $\dfrac{5\pi}{6}$ and $-\dfrac{\pi}{2}$ straight away; both angles are outside the principal ranges.